

Apache Parquet and Apache Arrow: Parquet is the *de facto* open columnar file format used by data lakes (S3, GCS, HDFS). It stores data in row groups with per-column encoding and compression. Arrow is the in-memory columnar format designed for zero-copy inter-process data sharing at CPU cache speed. Together they form the backbone of the modern data lakehouse stack.

Licence: Apache 2.0

URL: parquet.apache.org / arrow.apache.org

DuckDB: Often called ‘SQLite for analytics’, DuckDB is an embeddable, in-process columnar OLAP engine with no server required. It reads Parquet, CSV, and JSON files directly, integrates natively with Python and R, and uses a vectorised push-based execution model. A data scientist can run billion-row aggregations on a laptop in seconds.

Licence: MIT

URL: duckdb.org

Apache Druid: A high-performance, distributed, real-time analytics database designed for sub-second OLAP queries on event data. Druid ingests streaming data from Kafka in real time, stores it in compressed columnar segments, and serves queries across historical and real-time data simultaneously. Widely used for user-facing analytics dashboards.

Licence: Apache 2.0

URL: druid.apache.org

Apache Pinot: Originally developed at LinkedIn and open sourced in 2015, Pinot is a columnar OLAP store optimised for user-facing analytics at very low latency (single-digit milliseconds). It stores data in columnar segments with rich

index support: inverted, range, star-tree, bloom filter, and text indexes. Used at LinkedIn, Uber, and Walmart for real-time dashboards serving millions of end users.

Licence: Apache 2.0

URL: pinot.apache.org

Row-based and columnar databases are not competitors — they are complementary tools solving different problems. Row-based storage excels at the random, transactional access patterns of applications; columnar storage excels at the sequential, aggregation-heavy patterns of analytics.

The practical path forward for most organisations is a two-tier strategy: keep a battle-tested row-based database for transactions and invest in a modern columnar engine for analytics — connecting them with a reliable streaming or batch pipeline. This architecture scales to petabytes while keeping both workloads performant and cost-efficient. **END** 🐧

References

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